Series of real numbers

1) Find the sum:

a) $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+3)}$

b) $\sum_{n=1}^{\infty} \operatorname{arctg} \frac{1}{2n^2}$

a) $a_n = \frac{1}{n(n+1)(n+3)} = \frac{(n+2)}{n(n+1)(n+2)(n+3)}$

$= \frac{n+3-1}{n(n+1)(n+2)(n+3)} = \frac{1}{n(n+1)(n+2)} - \frac{1}{n(n+1)(n+2)(n+3)}$

$s_n = \sum_{k=1}^{n} a_k = \sum_{k=1}^{n} \left(\frac{1}{k(k+1)(k+2)} - \frac{1}{k(k+1)(k+2)(k+3)} \right)$

$= \underbrace{\sum_{k=1}^{n} \frac{1}{k(k+1)(k+2)}}_{S_1} - \underbrace{\sum_{k=1}^{n} \frac{1}{k(k+1)(k+2)(k+3)}}_{S_2}$

$S_1 = \frac{1}{2} \sum_{k=1}^{n} \left(\frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)} \right) = \frac{1}{2} \cdot \left(\frac{1}{2 \cdot 1} - \frac{1}{2 \cdot 3} + \frac{1}{2 \cdot 3} - \dots \right.$

$\left. + \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right) = \frac{1}{2} \left(\frac{1}{2} - \frac{1}{(n+1)(n+2)} \right)$

$$S_2 = \frac{1}{3}\sum_{k=1}^{n}\left(\frac{1}{k(k+1)(k+2)} - \frac{1}{(k+1)(k+2)(k+3)}\right) = \frac{1}{3}\left(\frac{1}{6} - \frac{1}{(n+1)\ldots}\right.$$

$$\lim_{n\to\infty} a_n = \frac{1}{4} - \frac{1}{18} = \frac{9-2}{36} = \frac{7}{36}$$

ftw: $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+3)(n+4)} = \frac{5}{144}$ rez

2) $\operatorname{arctg} x - \operatorname{arctg} y = \operatorname{arctg}\frac{x-y}{1+xy}$, $xy > -1$

$$a_n = \operatorname{arctg}\frac{1}{2n^2} = \operatorname{arctg}\frac{2}{4n^2} = \operatorname{arctg}\frac{2}{1+4n^2-1}$$

$$= \operatorname{arctg}\frac{(2n+1)-(2n-1)}{1+(2n-1)(2n+1)} = \operatorname{arctg}(2n+1) - \operatorname{arctg}(2n-1)$$

$$S_n = \sum_{k=1}^{n} a_k = \sum_{k=1}^{n}\left(\operatorname{arctg}(2k+1) - \operatorname{arctg}(2k-1)\right)$$

$= \operatorname{arctan} 3 - \operatorname{arctan} 1 + \operatorname{arctg} 5 - \operatorname{arctan} 3 + \ldots +$
$\operatorname{arctan}(2n+1) - \operatorname{arctan}(2n-1)$

$= \operatorname{arctan} - \frac{\pi}{4} \Rightarrow s = \lim_{n\to\infty} S_n = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$

3) $\sum_{n=1}^{\infty} \operatorname{arctg}\frac{2}{n^2+n+4}$

$\frac{2}{n^2+n+4}$

$\frac{2}{n(n+1)+4}$

$\frac{2}{4\left(\frac{n(n+1)}{4}+1\right)}$

$n^2+n+4 = n^2+n+3+1$ $\quad \frac{\frac{1}{2}}{\frac{n(n+1)}{4}+1} =$

$= (n$

$\frac{n+1}{2} - \frac{n}{2} = \Rightarrow \arctan\left(\frac{n+1}{2}\right) - \arctan\left(\frac{n}{2}\right)$

$\Rightarrow s_n = \sum_{k=1}^{n} \left(\arctan\left(\frac{n+1}{2}\right) - \arctan\left(\frac{n}{2}\right)\right)$

$\Rightarrow \quad s_n = \arctan\frac{2}{2} - \arctan\frac{1}{2} + \arctan\frac{3}{2} - \arctan\frac{2}{2}$

$+ \ldots + \arctan\frac{n+1}{2} - \arctan\frac{n}{2}$

$= \arctan\frac{n+1}{2} - \arctan\frac{1}{2} = \frac{\pi}{2} - \arctan\frac{1}{2}$

Remark: $\arctan x + \arctan\frac{1}{x} = \frac{\pi}{2}$

$\Rightarrow \arctan 2$

2) $\sum_{n=1}^{\infty} \{(2+\sqrt{5})^{2n-1}\}$

$[x] \qquad x = [x] + \{x\}$

$0 \le \{x\} < 1$

Let $x_n = (\sqrt{5}+2)^{2n-1} - (\sqrt{5}-2)^{2n-1} \in \mathbb{Z}$

$(a+b)^n = \binom{n}{0} a^n b^0 + \binom{n}{1} a^{n-1} b^1 + \ldots + \binom{n}{n} a^0 b^n$

$$x_n = \binom{2n-1}{0}(\sqrt{5})^0 2^{2n-1} + \binom{2n-1}{1}(\sqrt{5})^1 2^{2n-2} + \dots \binom{2n-1}{2n-1}(\sqrt{5})^{2n-1} 2^0 - \left[\binom{2n-1}{0}(\sqrt{5})^0 2^{2n-1} + \binom{2n-1}{1}(\sqrt{5})^0 2^{2n} - \dots + \binom{2n-1}{2n-1}(\sqrt{5})^{2n-1}(2)^0\right]$$

$$(\sqrt{5}+2)^{2n-1} = x_n + (\sqrt{5}-2)^{2n-1}$$

$$\{(\sqrt{5}+2)^{2n-1}\} = \{x_n + (\sqrt{5}-2)^{2n-1}\}$$

$$= \{(\sqrt{5}-2)^{2n-1}\} = (\sqrt{5}-2)^{2n-1}$$

$$(\sqrt{5}-2)^{2n-1} \in (0,1)$$

$$\Rightarrow \sum_{n=1}^{\infty} \{(\sqrt{5}+2)^{2n-1}\} = \sum_{n=1}^{\infty} (\sqrt{5}-2)^{2n-1}$$

$$= \underbrace{(\sqrt{5}-2) + (\sqrt{5}+2)^3 + \dots}_{\text{series geometrica}}$$

$$= (\sqrt{5}-2)\frac{1}{1-(\sqrt{5}-2)} =$$

Hw: $\sum_{n=1}^{\infty} \{(1+\sqrt{2})^{2n-1}\} = \frac{1}{2}$

2) Test for convergence

a) $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n!)} \cdot a^n$, $a>0$

$b_n = \frac{(n!)^2}{(2n!)} \cdot a^n$

$l = \lim_{n\to\infty} \frac{b_{n+1}}{b_n} = \lim \frac{(n+1)!^2}{(2n+2)!} \cdot a^{n+1} \cdot \frac{(2n!)}{n!^2 \cdot a^n}$

$\lim = \frac{(n+1)^2}{(2n+1)(2n+2)} \cdot a = \lim \frac{an^2+2an+a}{4n^2+6n+2} = \frac{a}{4}$

$\frac{a}{4} < 1 \Rightarrow a < 4 \Rightarrow \sum_{n=1}^{\infty} b_n$ is conv

$\frac{a}{4} > 1 \Rightarrow a > 4 \Rightarrow \sum_{n=1}^{\infty}$ is div

$a = 4 \Rightarrow$ Test fails

For $a=4 \Rightarrow b_n = \frac{(n!)^2}{(2n!)} \cdot 4^n$

Raabe duhamel $\Rightarrow n\left(\frac{a_n}{a_{n+1}} - 1\right)$

$= n\left(\frac{4n^2+6n+2}{4n^2+8n+4} - 1\right) = n\left(\frac{-2n-2}{4n^2+8n+4}\right)$

$= \frac{-2n^2-2n}{4n^2+8n+4} = -\frac{1}{2} < 1 \Rightarrow a=4 \Rightarrow \sum_{n=1}^{\infty} b_n$ is div

b) $x_n = \left(\frac{an+b}{cn+d}\right)^n$

$l = \lim \sqrt[n]{x_n} = \lim \frac{an+b}{cn+d} = \frac{a}{c}$

$\frac{a}{c} < 1 \Leftrightarrow a < c \Rightarrow \sum_{n=1}^{\infty} x_n$ is conv

$\frac{a}{c} > 1 \Leftrightarrow a > c \Rightarrow \sum_{n=1}^{\infty} x_n$ is div

$a = c \Rightarrow$ Test fails

$a=c \Rightarrow x_n = \left(\frac{an+b}{an+d}\right)^n$

$\lim x_n = \left(\left(\frac{b-d}{an+d}+1\right)^{\frac{an+d}{b-a}}\right)^{\frac{n(b-nd)}{an+d}}$

$= e^{\frac{b-d}{a}} > 0 \Rightarrow \sum_{n=1}^{\infty} x_n$ is divergent

c) $\sum_{n=1}^{\infty} \frac{a(a+1)\dots(a+n-1)}{n!} \cdot \frac{1}{n^\alpha}, \; a > 0, \alpha \neq 0$

$l = \lim \frac{a(a+1)\dots(a+n)}{(n+1)!\,(n+1)^\alpha} \cdot \frac{n!\,n^\alpha}{a(a+1)\dots(a+n-1)}$

$$= \frac{a+n}{(n+1)} \cdot \left(\frac{n}{n+1}\right)^{\alpha} = \frac{n^{\alpha+1} + a n^{\alpha}}{(n+1)^{\alpha+1}} = 1$$

$$\lim n \left[\frac{n+1}{a+n} \left(\frac{1}{n}+1\right)^{\alpha} - 1 \right]$$

$$\lim n \left[\frac{n+1}{a+n} \left(1+\frac{1}{n}\right)^{\alpha} - \frac{n+1}{a+n} + \frac{n+1}{a+n} - 1 \right]$$

$$= \lim n \left[\frac{n+1}{a+n} \left(\left(1+\frac{1}{n}\right)^{\alpha} - 1 \right) + \frac{1-a}{a+n} \right]$$

$$= \lim_{n \to 1} \frac{n}{a+n} \left(\frac{\left(1+\frac{1}{n}\right)^{\alpha} - 1}{\frac{1}{n}} \right) + \frac{n(1-a)}{n}$$

$$= \lim \frac{\alpha (n+1)}{a+n} + \frac{n(1-a)}{n} = \alpha + 1 - a$$

$\alpha < a \Rightarrow \sum b_n$ in α

$\alpha > a \Rightarrow \sum b_n$ in conv